John Filhmar Ola

Full Stack Software Engineer

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PROFESSIONAL SUMMARY

Full-stack engineer who owns systems end to end, from architecture and TypeScript development across React and Node.js to CI/CD and the servers they run on. Sole developer of production systems with real users, including an encrypted messaging app on iOS and Android and a civic platform live in three cities. Uses Claude Code daily to ship faster, with every generated change reviewed and tested. Promoted from Developer to Lead IT within the first year.

PROFESSIONAL EXPERIENCE

Lead IT Systems Developer | YouCode Technologies Corporation
July 2025 - Present

- Promoted to Lead IT within the first year; own six commercial systems end to end across architecture, development, testing, CI/CD and production operations.
- Built TypeScript microservices over gRPC and RabbitMQ with per-service PostgreSQL and MongoDB, Redis caching, and WebSocket delivery fanned out across gateway instances.
- Own CI/CD for the product suite with automated tests, SAST, SCA, container and secret scanning, and signed containerized deploys; run Prometheus and Grafana monitoring and resolve live incidents.

Frontend and Web Developer Intern | Ynnovision Enterprise Solutions
February 2025 - May 2025

- Built responsive front-end features across production web apps during a 486-hour internship, turning design mockups into cross-browser UI and fixing interface defects.

PROJECTS

Tara LEVEL UP! | tara.filhmar.online
On-Device LLM, Offline Inference, Multimodal AI (image/voice/text), Android
Problem: Habit and motivation apps lean on the honor system or cloud AI that needs a live connection. **Solution:** A gamified quest app where a local LLM verifies each finished task from a photo, voice or short quiz entirely on-device, awards XP under fixed rules, and recomputes the leaderboard server-side to stop cheating. **Result:** Built from scratch in 21 hours at the AppBuildersPH Hackathon 2026, running multimodal AI fully offline on the phone with no signal required.

USAPP | usapp.ph
React Native (Expo), NestJS, gRPC, RabbitMQ, Socket.IO, Redis, PostgreSQL, MongoDB
Problem: Users needed private messaging where the server itself cannot read message content. **Solution:** End-to-end encryption with client-side keys, passwordless OTP sign-in with revocable device sessions, WebSocket delivery coordinated through Redis, and a partner API for outside organizations. **Result:** Live on iOS and Android, and adopted as the push-notification backbone for every other company product.

Real-Time Trade Blotter | fusion.filhmar.online
TypeScript, React, Express, PostgreSQL, Socket.IO, Redis, Docker, AWS EC2, Claude Code
Problem: A trading platform assessment needed a real-time blotter where every trade booking, amendment and cancellation reaches all users instantly and correctly. **Solution:** Built it in about a week with Claude Code, documented in an AI usage report and prompt log, with exact decimal prices, a database-enforced audit trail, optimistic concurrency and keyset pagination. **Result:** Deployed to AWS behind nginx with TLS, with 325 unit, 39 integration and 10 end-to-end tests and a load-test p95 of 209 ms.

JMS E-Talyer | jms-e-talyer.shop
Next.js, Express, MongoDB, Redis, Docker, GitHub Actions
Problem: Multi-branch automotive shops needed shared inventory, sales and service records that stay accurate even when a branch goes offline. **Solution:** An offline-capable PWA with barcode scanning and server-authoritative conflict resolution that replays offline orders safely on reconnect. **Result:** Live in production with 400+ backend tests and security scanning gating every release, on a server I operate.

YouFlow | kanban.youcodetech.com
Express, Next.js, Prisma, PostgreSQL, Redis, Model Context Protocol
Problem: Teams working with AI agents had no safe way to let agents act on their work. **Solution:** An MCP server where each agent acts with one user's exact permissions, AI estimates held behind a supervisor approval gate, and an append-only audit log of every action. **Result:** Agents do real work while unverified numbers never reach management reports.

YouGuard Master Bridge Server | Deployed on customer sites
Go, MQTT, WebSocket, WebRTC, RTSP, PostgreSQL, MongoDB, React
Problem: Attendance and security events had to survive site network outages, and two separate bridges each handled only one hardware type. **Solution:** One Go service with per-hardware-family adapters that writes every event to disk before sending and replays unacknowledged events oldest first on reconnect. **Result:** Events are buffered through outages instead of lost, one integration path for all hardware, and RTSP camera feeds streamed to a live WebRTC video wall.

CORE STRENGTHS

- End-to-end ownership, from architecture to production
- Real-time and event-driven systems
- Durability and failure recovery by design
- Security automation built into CI/CD
- AI-assisted delivery with reviewed, tested output

AI-ASSISTED DEVELOPMENT

- Claude Code daily for building, testing, debugging and documentation
- Project context files and custom skills keep the AI on locked-in decisions
- Every generated diff reviewed, with tests as the gate before merge
- Ships faster: a full real-time trading app with 325 unit tests in about a week

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Go, Java, C#, SQL, Bash

Front End: React, Next.js, React Native (Expo), Tailwind CSS, TanStack Query, PWAs

Back End: Node.js, Express, NestJS, Python (Django), REST APIs, gRPC, WebSockets, Socket.IO

Messaging: RabbitMQ (AMQP), Redis Pub/Sub, Background Jobs

Data: PostgreSQL, MySQL, MongoDB, SQLite, Redis, Prisma, pgvector

DevOps: Docker, GitHub Actions, CI/CD, Linux, nginx, AWS EC2, Prometheus, Grafana

Security: SAST, SCA, DAST, Trivy, gitleaks, cosign, SOPS

AI Features: LLM APIs, MCP Servers, Prompt Engineering, Human Review Gates

Testing: Jest, Vitest, Playwright, JUnit, k6 Load Testing

CERTIFICATIONS

AppBuildersPH Hackathon 2026
App Builders PH, SM Makati, October 2026

- Built a local-AI mobile app in 21 hours with team BOOM TARA TARA G

Programming (Java) NC III
TESDA, BCRV Tech-Voc, 2025

- Top 1 Awardee of the cohort
- Best in Participation and Best in LMS Submission
- Graduation Inspirational Speaker
- Capstone: Healthy Gram, a Java desktop app built on OOP principles

EDUCATION

BS Information Technology
Mindoro State University, 2021 - 2025

- Capstone: Kalusugapp, a patient records and appointment system deployed at the Victoria Municipal Health Office

LANGUAGES

English (professional working), Filipino and Tagalog (native)